

An Investigation Into Dropout Indicators in Secondary Technical Education Using Explainable Artificial Intelligence

Francisco da Conceição Silva^{1b}, Andre Macedo Santana, and Rodrigo Miranda Feitosa

Abstract—With the increasing application of Artificial Intelligence (AI) in education, it becomes essential to understand the factors influencing school dropout, as the educational context demands reliable decision-making. This study investigates dropout indicators in secondary-level technical courses at a Brazilian institution, using Explainable AI (XAI) techniques applied to Machine Learning models. The study analyzed data from 15,084 students to identify the main factors contributing to school dropout, utilizing predictive models and applying the explainability techniques LIME and SHAP to highlight key dropout factors, thereby improving prediction transparency. The results show that decision tree-based models performed best, with Random Forest achieving 85% Recall, effectively identifying students at risk of dropout. LIME and SHAP consistently highlighted school attendance, family income, and place of residence as key dropout factors. The analysis of the explainers showed that students with low attendance and lower income are more likely to drop out. These findings highlight the importance of targeted educational policies, such as scholarship programs, transportation assistance, and personalized academic support, especially for vulnerable students. This study contributes to the understanding of factors associated with school dropout and provides insights for the formulation of more effective educational policies. Strategies such as scholarship programs, transportation assistance, and academic monitoring can be implemented to reduce dropout rates.

Index Terms—Explainable artificial intelligence (XAI), machine learning, educational data mining, dropout indicators.

I. INTRODUCTION

SCHOOL dropout is a recurring issue in various educational institutions around the world, whether the mode of delivery is in-person or distance learning [1]. The high dropout and failure rates are concerning, as various factors—such as personal, academic, physical, familial, and social issues—can negatively impact student performance, leading to failure or dropout [2].

In Brazil, the vocational education model for high school students or those who have already completed it is called the technical high school course. This model can be offered in two

ways: integrated with high school (either as part of the course or concurrently) or after its completion [3].

The National Household Sample Survey (PNAD - 2022) [3] showed that the main cause of school dropout is the need to study (40.2%), followed by lack of interest (24.7%). It is important to note that, for women, pregnancy (22.4%) and household chores (10.3%) are significant factors. This issue is often worsened by socioeconomic challenges, such as low family income and limited access to transportation, especially in rural areas.

This not only affects individual futures but also limits the country's economic potential. Explainable Artificial Intelligence (XAI) offers a promising approach to understanding the complex factors behind dropout, allowing stakeholders to make informed and transparent decisions.

In vocational education, dropout rates vary depending on the mode of instruction. Data from the Nilo Peçanha Platform reveals that in 2023, the dropout rate for integrated high school was 8.26%, while in Proeja subsequent courses (National Youth and Adult Education Program), the rate reached an alarming 46.79%.

Concomitant and subsequent technical courses registered dropout rates of 29.97% and 22.98%, respectively. These numbers indicate the need for specific policies for each mode to reduce dropout and promote student retention [4].

These data highlight a shared concern among educational institutions, educators, coordinators, and society regarding high student dropout rates in schools [1], [5].

Studies [6], [7], and [8] show how computational tools leveraging Artificial Intelligence (AI) and Explainable AI (XAI) enhance the interpretability of Machine Learning (ML) models, aiding administrators and educators in mitigating school dropout and improving the teaching-learning process.

ML models can be used for a variety of tasks, such as predicting student performance, identifying students at risk of dropout, and personalizing education. Although the application of AI in education is significant and desirable, it is important to expand the discussion on the explainability of results obtained by ML models [9], so that the explainability of AI systems in the educational domain provides reliability and better understanding for stakeholders [10], [11], [12].

Thus, the aim of this study was to investigate the identification of dropout indicators for technical high school students through the application of XAI approaches on data from a teaching institution.

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For this purpose, several actions were taken: a bibliographic study to expand understanding of dropout in technical education; investigation and definition of ML techniques used in the development of predictive models for identifying dropout indicators; and application of XAI techniques to explain the results obtained by these models.

In terms of advancing knowledge on the topic, this study offers several potential contributions: providing an overview of the use of explainable models that favor the reliable identification of dropout indicators; highlighting the most used XAI techniques in this context; identifying the limitations of using explainability in ML models; and providing valuable insights to educators and administrators, helping to improve education and student performance.

II. LITERATURE REVIEW

A. Explainability of ML Models

AI explainability refers to the methods and techniques applied to systems that aim to enhance human understanding of the results and decisions generated by ML models. This definition arises from the intrinsic operational complexity of many of these methods, making it difficult to understand or explain the steps taken to achieve a specific result [13].

The use of XAI in ML models is a rapidly growing research area aimed at making algorithms more understandable, which is crucial for building trust in their results.

According to [10], interest in XAI goes far beyond the need for explainable models, transcending into legal issues of user data protection. In Brazil, the General Data Protection Law (LGPD) guarantees users the right to clear, accurate, and accessible information about the processing of their data, under penalty of legal sanctions such as fines and suspension of the activities of the organization that uses this data [14].

B. Xai Techniques

There are various XAI techniques available, each with its own well-defined type and scope of action, which constitute the approaches to providing explanations of ML models. These approaches can be categorized into three main dimensions: method, stage, and scope, as described in the literature [10], [15], and [16].

The method dimension refers to the approach used to generate ML models, including interpretable glass-box models and complex closed-box models.

The stage dimension defines when explanations occur: ante-hoc explanations are generated during training (e.g., rule-based systems), while post-hoc explanations are applied after training to interpret closed-box models.

Finally, the scope dimension determines the breadth of the explanation. Local explanations focus on the model's behavior for a specific observation, simulating the operation of the closed-box model on similar samples. Global explanations, on the other hand, aim to understand the model's overall behavior across all expected scenarios, providing a comprehensive view of its decision-making process.

This structured categorization of XAI techniques provides a clear frame of study for understanding how different approaches

can be applied to enhance the interpretability of ML models, particularly in educational contexts.

Some of the most popular tools for studying with XAI are: Local Interpretable Model-agnostic Explanations (LIME) [17], Shapley Additive exPlanations (SHAP) [18], and ELI5.

SHAP and LIME are model-agnostic tools that generate explanations of why the models' predictions are correct and are of the post-hoc type [13]. LIME provides local explanations, while SHAP provides both local and global explanations, and is compatible with any type of method, though it is often used with closed-box models.

LIME is a method that explains individual predictions of complex models by approximating them with simpler, interpretable models. Some of the most important features of LIME are its accessibility and simplicity [17].

The central idea behind LIME is that it can provide explanations for any supervised learning model, treating it as a 'closed box' separately.

The objective is to develop an interpretable model using observations generated by LIME, which are based on similar cases with minimal parameter changes to produce variations. The model is then built using this set of observations generated by LIME.

SHAP is a method rooted in game theory, which uses Shapley values to quantify the average marginal contribution of each variable to the model's predictions [8], [18].

The Shapley value represents the fair and proportional contribution of each parameter to the overall prediction, analogous to how players in a game contribute to the total outcome.

In the context of a student dataset, SHAP treats each attribute as a 'player,' assessing its individual impact on predicting a student's outcome, such as dropout risk or academic performance.

C. Using Xai to Tackle School Dropout

In the context of education, XAI can be a valuable tool for preventing school dropout. Applying XAI in this scenario can enable a better understanding of the factors that lead to dropout and thus make it possible to implement more effective strategies to combat it.

In addition, XAI can help ensure that the AI systems used in education are studying as expected, which may be necessary to meet regulatory standards, or may be important to allow those affected by a decision to challenge it.

Explainable ML models and XAI techniques play a crucial role in addressing dropout by making predictions more transparent. By making ML algorithms more understandable to humans, it is possible not only to build more reliable AI systems but also to develop more effective strategies to keep students engaged and reduce dropout rates.

III. RELATED STUDIES

This section presents some related studies relevant to the purpose of this research and that are crucial for a better understanding of XAI approaches.

The research conducted by Melo et al. [6] focuses on predicting school dropouts at the Federal Institute of Rio

Grande do Norte (IFRN) using XAI. The study achieved 99% accuracy in training and 97% in testing. The authors proposed a new explainability index (which measures the effectiveness of XAI methods used in dropout prediction) to evaluate XAI framestudies, with LIME meeting 8 of the 14 established metrics. It was identified that the race “parda” (mixed race) did not contribute to the dropout prediction. However, the study has limitations, including the use of a single dataset and the lack of evaluation of the proposed index in other contexts.

Addressing the challenge of student retention, Pei and Xing [7] presents an approach to identify at-risk students, focusing on the explainability of output and weekly analysis of learning activities. It uses methods such as Random Forest, Decision Tree, Support Vector Machine, and a model proposed in the study. The performance of the models is evaluated through weekly AUCs and average performance (precision, recall, and F1-score) over each course. The study adopts a Top-K metric to examine the number of at-risk students. The model can identify with high accuracy weekly, providing specific details on the model’s best performance for each dataset used.

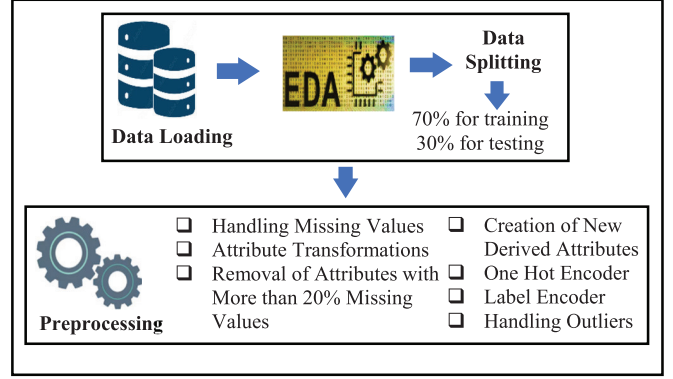
Alwarthan et al. [8] proposes an explainable model to identify at-risk students in higher education using various data mining algorithms. The Random Forest algorithm had the best performance with an accuracy of 0.89. Explanation techniques such as LIME and SHAP were used for analysis, and the most influential features were high school GPA, number of absences, and the number of failed courses. However, the study only addresses the prediction of at-risk students in the preparatory year and does not consider the impact of external factors on student performance.

Nagy and Molontay [19] investigated the use of ML and XAI in predicting dropout in higher education, highlighting the effectiveness of CatBoost in identifying at-risk students based on pre-leveling performance variables, such as high school graduation exam scores. The authors emphasize the role of SHAP and LIME in the interpretability of predictions, allowing for personalized academic interventions. They also point out challenges in understanding these tools and suggest improvements in communicating the results.

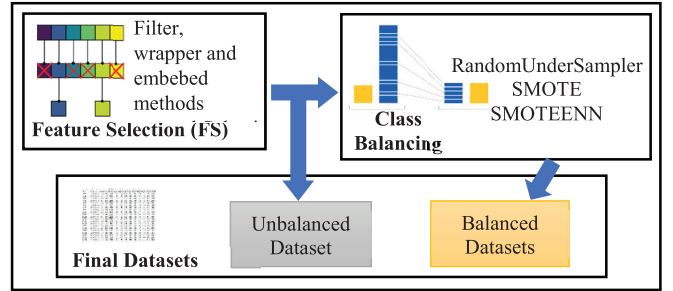
Zanellati et al. [20] compare the performance of Random Forest (RF) and Feature Tokenizer Transformer (FTT) in predicting dropout, using a dataset of more than 40,000 students from an Italian university. The results indicate that credits (ECTS) are a central factor in prediction, with FTT outperforming RF in predictive accuracy. Despite the model’s robustness, the authors highlight the limitation of approaches solely based on quantitative metrics and suggest the inclusion of qualitative data for a broader understanding of the dropout phenomenon.

Krüger et al. [21] present an explainable machine learning model to predict school dropout based on academic and socioeconomic data from 19 Brazilian schools. With an AUC-PR of up to 89.5% and accuracy of up to 95%, the study highlights that challenges vary depending on the educational level, being more pronounced in early childhood education. In addition to predicting dropout, the model identifies underlying factors contributing to abandonment, enabling proactive interventions.

Step 1 – Data Loading and Preprocessing



Step 2 – Feature Selection and Class Balancing



Step 3 – Application of XAI in ML Models

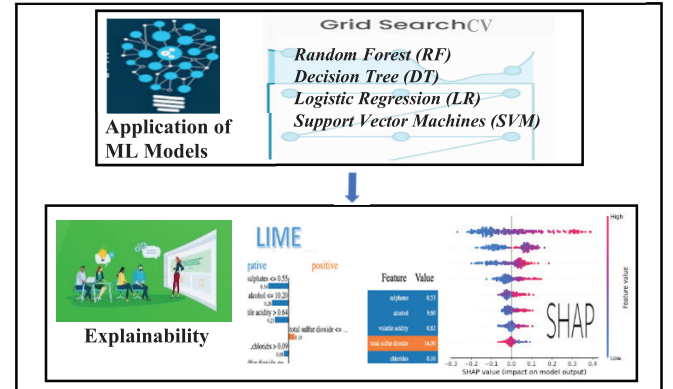


Fig. 1. Flow of the methodological pipeline for the application of XAI on student data, including preparation, balancing, feature selection, model training, and XAI.

In summary, the following approaches, essential in this study, are distributed across these and other studies as follows: XAI techniques: all; validation metric: cross-validation [7], [8], [21], [22], cohort approach [19]; feature selection: [6], [8]; class balancing: [6], [7], [8], [20]; XAI techniques applied to more than one ML model: none.

IV. MATERIALS AND METHODS

This section describes the methodological procedures adopted for conducting experiments in this study. The pipeline developed for this methodology, the dataset, and experimentation reports are available at <https://bit.ly/case-study-dropout-xai>.

Fig. 1 presents the steps of the pipeline for conducting the experiments in this study.

TABLE I

DISTRIBUTION OF 15,084 STUDENTS IN SECONDARY-LEVEL TECHNICAL EDUCATION BY COURSE TYPE, WITH INTEGRATED COURSES BEING THE MOST COMMON

Type of course	Total number of students
Integrated	10,070
Subsequent	3,902
Concomitant	718
Integrated EJA (Youth and Adult Education)	394

TABLE II

SUMMARY DESCRIPTION OF THE MAIN DATASET VARIABLES, PROVIDING AN OVERVIEW OF THE CHARACTERISTICS OF THE ANALYZED DATA

Category	Attributes (some highlighted)	Type
Personal Information	Gender (M/F), Race (categories), Marital Status, Age (15-29 years), Number of Children (0-10), Type of Residential Area (urban/rural)	Categorical / Numerical
Academic Information	Campus, Course, Year of Entry (2010-2022), Entry Period (1 or 2), Mode, Attendance Percentage (37%-100%), Failures (0-5)	Categorical / Numerical
Socioeconomic Information	Time Spent Out of School, Gross Income (R\$), Household Company, Mother's Education Level, Father's Education Level, Studies (Occupation)	Categorical / Numerical
Student Status	Status (0 = Completed, 1 = Dropped Out)	Categorical

A. Data Understanding and Pre-Processing

The dataset used in this study comes from secondary-level technical courses at a public educational institution and consists of personal, academic, and socioeconomic data, totaling 15,084 student records and 31 columns (30 predictive attributes +1 class attribute). The distribution of students across different types of courses is presented in Table I.

Table II describes the original attributes of the dataset. Although it lists 20 attributes, all 31 are represented, as the application of OneHotEncoder on gender, race, and marital status generated 14 new derived attributes, replacing the original ones. The complete list can be found in the supplementary material.

Analyzing the target class of the dataset, which corresponds to the attribute (variable) 'situation' (Student Status), it was found that a student could fall into various situations that, in reality, reflect the case of passing or dropping out of a course. Therefore, the target class was transformed into a binary class, whose values were derived as follows:

- Completed [Class 0]: derived from the value 'Completed';
- Dropped Out [Class 1]: derived from the values 'Dropout', 'Cancelled', 'Compulsory Cancellation', 'Jubilated', 'Externally Transferred'.

Although the target class was reduced to a binary class, the class distribution generated an imbalance in the proportion of 78% (11,703 records) for the value 'Completed' and 22% (3,381 records) for the value 'Dropped Out', considering the original (unbalanced) data. Section IV-B of this paper addresses the analysis and treatment of class imbalance.

TABLE III

NUMBER OF INSTANCES BEFORE AND AFTER CLASS BALANCING, DETAILING THE APPROACHES ADOPTED AND THE IMPACT ON DATA DISTRIBUTION. A/SEPARATES THE TRAINING AND TEST DATA

Class	Unbalanced	Subsampling (sub)	Oversampling (super)	Combination (sub/super)
Dropped Out	2,408 (23%) / 973 (22%)	2,408 (50%) / 973 (50%)	8,150 (50%) / 3,553 (50%)	5,937 (57%) / 2,501 (58%)
Completed	8,150 (77%) / 3,553 (78%)	2,408 (50%) / 973 (50%)	8,150 (50%) / 3,553 (50%)	4,466 (43%) / 1,773 (42%)

Attributes indicating student identification were removed, as well as those with a percentage of null values above 20%, resulting in a manual selection of attributes. Outliers were treated using the Interquartile Range (IQR), with the removal of extreme values. For categorical variables, OneHotEncoder and LabelEncoder encoding were applied, while numerical variables were kept without normalization.

In addition, the experiments were conducted using the following feature selection (FS) methods: filter methods, wrapper methods, and embedded methods.

For each FS method, a technique was used, as follows:

- Filter methods: ANOVA
- Wrapper methods: Backward Selection
- Embedded methods: Feature Importance

The above techniques were used to provide methodological diversification without prioritizing the superiority of one technique over another. Techniques such as genetic algorithms and RFE, for example, were tested but not included in this study. Future studies could explore a wider range of methods to assess their impact on explainability and predictive performance.

B. Analysis of Class Imbalance and Creation of Balanced Datasets

Class imbalance in a dataset represents a problem for defining the performance of a model in terms of its ability to make correct predictions.

Table III presents the balanced and unbalanced datasets used in the experiments. To deal with class imbalance, subsampling (RandomUnderSampler), oversampling (SMOTE [23]), and a hybrid approach (SMOTEENN [24]) were employed. SMOTE was chosen because it generates synthetic examples without direct replication, reducing overfitting, while SMOTEENN combined oversampling with noise removal using Edited Nearest Neighbors (ENN).

We selected these techniques based on the literature [6], [7], and [8] and the need to maintain data interpretability. SMOTE was preferred over techniques such as ADASYN because it generates fewer examples close to outliers, and SMOTEENN was adopted over Tomek Links and SMOTETomek due to its better performance in removing noise. In addition, one

technique per approach was chosen, balancing applicability and interpretability of the results.

C. Application and Comparison of Classification Algorithms and Model Hyperparameter Selection

We conducted the experiments using the following algorithms:

- Random Forest (RF): Builds multiple decision trees with random attributes, generalizing results and minimizing the risk of overfitting [25], [26];
- Decision Tree (DT): Partitions the data based on division mechanisms, evaluating class purity using the Gini Index [27]; widely applied in classification problems in education [25];
- Logistic Regression (LR): Probabilistic statistical model, mainly used for classification but also applicable to regression [28];
- Support Vector Machines (SVM): Classifies categories using hyperplanes that maximize the margins between classes [27], [29].

The RF and DT models were chosen for their efficiency in handling unbalanced data and their widespread use in the literature on school dropout [6], [7], [8], [20], which is essential for the application of XAI techniques. SVM and LR were used as baselines for comparison.

More complex models, such as Gradient Boosting and Neural Net, were discarded due to the high computational cost and the increased number of experiments conducted in the study, although they could be explored in the future to balance performance and explainability.

To ensure that the ML models had the best performance without compromising processing time, hyperparameter tuning was performed using the GridSearchCV technique. This technique searches the grid to find the hyperparameters that best fit the model for classification.

The performance of the models was evaluated using standard metrics, including precision (the proportion of true positives among predicted positives), recall (the proportion of true positives correctly identified), F1-score (the harmonic means of precision and recall), AUC (the area under the ROC curve), and accuracy (the proportion of correct predictions). In addition, 10-fold cross-validation [30], referred to as MEDIA_10_CV, was applied, representing the average performance of the model across 10 subsets of the data, aiming to verify the robustness of the model for predicting school dropout.

To ensure that only the models with the best performance remained in the more detailed discussion of the results, the following selection criteria were defined, precisely in the following stages: 1st) highest Recall; 2nd) highest Precision; 3rd) highest MEDIA_10_CV; and 4th) highest values of the other metrics defined in this study.

D. Application of Shap and Lime Explainability Techniques to ML Models

The choice of ML techniques involves a balance between generalization capacity and explainability [31]. Simple models,

such as LR and DT, are intuitive but can be limited in scenarios with many parameters. More complex methods, such as RF, SVM, and neural net studies, are effective but function as “closed boxes,” with no clear explanation of the results [32].

To overcome this limitation of “closed-box” models, approaches focusing on AI explainability have been created to help explain predictions. The most common XAI techniques are: LIME [17] and SHAP [18], which are adopted in this study.

LIME and SHAP were chosen due to their flexibility, widespread adoption in the literature and XAI community [33], and suitability for a variety of ML models, facilitating the reproduction and validation of results.

Compared to alternatives such as ELI5 and Anchors, which have limitations in complex models or depend on discretization, LIME and SHAP demonstrated greater scalability and suitability to the context of the study, ensuring consistent and flexible explanations.

E. Scenario for the Experiments

An experiment was conducted applying the techniques discussed in Section II, as follows:

- FS Techniques: Filter Method (ANOVA), Wrapper Method (Backward Selector), and Embedded Method (Feature Importance);
- Data Samples: Unbalanced data and balanced data (subsampling, oversampling, and a combination of subsampling and oversampling);
- ML Algorithms: RF, DT, LR, and SVM;
- XAI Methods: LIME and SHAP.

The original dataset consists of 15,084 records and 31 columns, defined here as unbalanced data (dnb). The balanced datasets were generated from this dataset.

Considering that there are three FS techniques, four ML models, and four types of datasets, the experiments consisted of 48 runs ($3 \times 4 \times 4$) of model training and validation, with the XAI methods being applied to each of the models. The aim was to identify the best combination of techniques and methods to create models with the best performance and, consequently, promote more significant analyses of the explainability of the selected models.

In each experiment, the hyperparameters were adjusted to best fit the data, evaluation metrics were calculated, and the model’s explainability was assessed.

The following techniques were used to implement the stages of the methodological process: Python programming language and libraries such as Scikit-learn, Pandas, Numpy, among others. The execution environment adopted in this study was the Google Colab platform, which is a hosted Jupiter Notebook service that requires no configuration to use and provides free access to computing resources, including GPUs and TPUs.

V. RESULTS AND DISCUSSION

A. Analysis of ML Model Performance

An experiment was conducted on different types of datasets and the model selection criterion defined in Section IV-C, was applied to identify the top five best-performing models.

TABLE IV

PERFORMANCE OF THE TOP FIVE ML MODELS IN THE EXPERIMENTS, EVALUATED USING METRICS SUCH AS PRECISION, RECALL, AND F1-SCORE, HIGHLIGHTING THE MODELS SELECTED FOR THE APPLICATION OF XAI TECHNIQUES

FS Method	Backward Selection	ANOVA	Backward Selection	ANOVA	Backward Selection
Model	RF	RF	RF	RF	DT
Balancing Type	Sub & Super	Sub & Super	Sub	Sub	Sub & Super
MEAN 10 CV	0.87	0.88	0.84	0.84	0.82
Precision	0.847	0.836	0.833	0.824	0.81
Recall	0.854	0.84	0.827	0.82	0.818
F1-score	0.849	0.838	0.827	0.82	0.811
Accuracy	0.852	0.841	0.827	0.82	0.813
AUC	0.854	0.84	0.827	0.82	0.818
(> Recall, > Precision, etc.)	1st	2nd	3rd	4th	5th

The results revealed that all models selected by the criterion were based on decision trees. These models were obtained using the Backward Selection and ANOVA methods on datasets balanced by oversampling, as well as by combining undersampling and oversampling techniques.

Thus, Table IV presents the performance of the top five classifiers evaluated in the experiments. The rows highlighted in gray indicate the two models to which explainability techniques will be applied. Based on the results obtained, the following analyses can be made.

The results indicate that the RF model, with SMOTEENN, achieved a Recall of 0.854, highlighting its effectiveness in identifying students at risk of dropout. The DT model had a slightly lower Recall (0.818), but it remains acceptable. Additionally, the combination of undersampling and oversampling improved the balance between precision and recall, reducing bias towards the majority class. These findings reinforce the importance of balancing strategies in predicting student dropout.

The Recall metric was prioritized because it is crucial for minimizing false negatives, ensuring that fewer at-risk students go unnoticed. This is particularly relevant in the educational context, where overlooking vulnerable students may result in missed opportunities for early intervention.

The superiority of the RF model can be attributed to its ability to handle complex and non-linear data, while the DT model, although less accurate, offers greater interpretability.

Models trained with the combination of undersampling and oversampling (SMOTEENN) exhibited a better balance between precision and recall, minimizing bias towards the majority class (students who completed the course). This balancing approach was essential to ensure that the model was not biased in favor of the majority class, allowing for more accurate identification of at-risk students.

Table V presents the attributes with the greatest impact on classification in the selected models, along with the tuned hyperparameters.

The hyperparameters tuned for the ML models, as presented in Table V, were the same for both RF and DT. Other hyperparameters, such as $n_estimators$ for RF and the number

TABLE V

SELECTED ATTRIBUTES AND TUNED HYPERPARAMETERS FOR THE RF AND DT MODELS, APPLIED TO DATA BALANCED BY COMBINING UNDERSAMPLING AND OVERSAMPLING, USING THE BACKWARD SELECTION FS METHOD

Selected Attributes	Best Parameter Set
'campus', 'Year of Entry (<i>anoingresso</i>)', 'Gross Income (<i>rendabruta</i>)', 'Modality (<i>modalidade</i>)', 'Age (<i>idade</i>)', 'Type of Residential Area (<i>tipo_area_residencial</i>)', 'Employment Status (<i>trabalha</i>)', 'Attendance Percentage (<i>percentual_frequencia</i>)', 'Failures (<i>reprovacoes</i>)', 'Gender Female (<i>genero_F</i>)', 'Marital Status Single (<i>estado_civil_Solteiro(a)</i>)', 'Entry Period (<i>periodoingresso</i>)', 'Race (<i>raca_3</i>)', 'Marital Status Not Declared (<i>estado_civil_Não_declarado</i>)', 'Marital Status Widowed (<i>estado_civil_Viúvo(a)</i>)', 'Course (<i>curso</i>)', 'Household Companionship (<i>companhia_domiciliar</i>)', 'Mother's Education Level (<i>mae_nivel_escolaridade</i>)', 'Number of Children (<i>qtd_filhos</i>)'	max_depth: 10, min_samples_leaf: 5, min_samples_split: 10

of trees, were kept at their default values. The application of these adjustments resulted in improved performance evaluation metrics for the models.

For the application of XAI methods, the RF and DT models will be used. The objective is to verify whether, despite the difference in performance, the explainability of these models is similar or presents significant differences regarding the reasons that led to a student's dropout.

In Section V-D, the practical implications of the analyses conducted above will be discussed, including: the difference in model performance for use in educational contexts; how these results can be applied to develop student retention strategies; and the importance of data balancing for obtaining more consistent results.

B. Explainability of Models With Lime

The explanations generated using LIME were conducted to assess the contribution of each attribute to the prediction outcome, as well as to determine which attributes had the greatest impact on the prediction result.

The LIME visualization consists of three parts: (1) Prediction probabilities for the binary class (Dropped Out/Completed); (2) Distribution of attributes by coefficient, indicating their positive (dropout) or negative (non-dropout) influence; and (3) A table of actual values for the attributes used in the explanation. The colors blue and orange represent "Completed" and "Dropped Out," respectively.

Fig. 2, generated by LIME for the RF model, provides a visual analysis of a single sample, highlighting the top five attributes that had the greatest influence on the model's classification of a student as "Dropped Out" from a course.

In This Case, the Prediction Probability of the Sample Was 83% for the 'Dropped Out' Class. Among the Top 5 Attributes With the Highest Impact on This Classification, LIME Identified That Attendance Percentage, Number of Failures, and Type of Residential Area (Rural) Had a Positive Influence on the Student's Dropout, While Gender (Male) and Household Composition (Living With Parents) Indicated

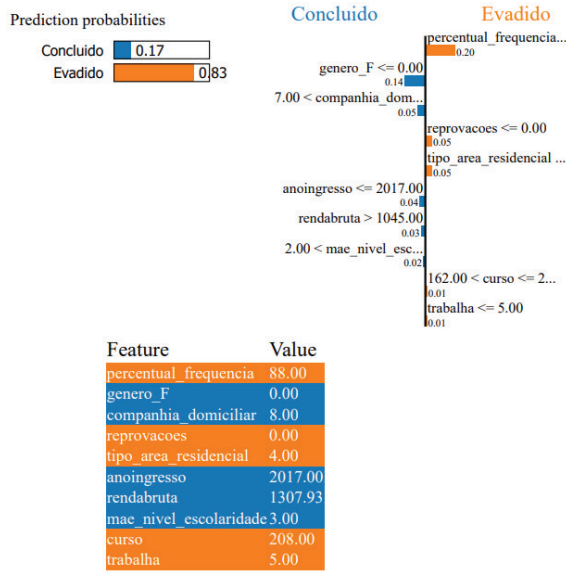


Fig. 2. LIME explanation for the RF model on a SMOTEENN-balanced sample, classified as 'Dropped Out' (83%). Attendance percentage was the most positively influential attribute (20%) in this classification.

TABLE VI

EXPLANATION OF THE ATTRIBUTES OF THE SAMPLE CLASSIFIED BY THE RF MODEL AS 'DROPPED OUT'

Attribute	Value	Explanation
Attendance Percentage (<i>percentual_frequencia</i>)	88	The student had an 88% attendance rate in the course; this attribute had a strong positive impact and was the most influential factor in the dropout prediction, contributing 20%.
Gender (<i>genero_F</i>)	0	The student is male; this had a strong negative impact and was the second most influential factor in the prediction, contributing 14%.
Household Companionship (<i>companhia_domiciliar</i>)	8	The student lives with their parents; this attribute had a negative impact and was the third most influential factor in the prediction, contributing 5%.
Failures (<i>reprovacoes</i>)	0	The student had no course failures; despite this, the attribute had a positive impact and was the fourth most influential factor in the prediction, contributing 5%.
Type of Residential Area (<i>tipo_area_residencial</i>)	4	The student is from a rural area; this attribute had a positive impact and was the fifth most influential factor in the prediction, contributing 5%.

a Negative Contribution, Meaning That the Student Is Not at Risk of Dropping Out.

Table VI presents the explanation for the 5 attributes that had the greatest impact on the student's situation as defined by the RF model.

For the DT model applied to the SMOTEENN-balanced sample classified as 'Dropped Out', LIME revealed the following key insights: the model's prediction probability was 84%. Gross income (9%), attendance percentage (9%), and number of failures (3%) positively influenced dropout prediction. Male gender (12%) and year of enrollment (5%) had a negative

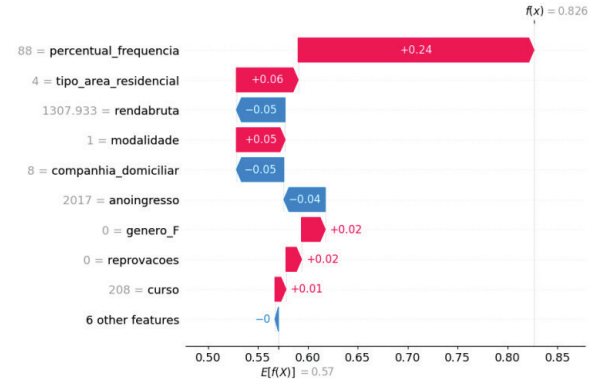


Fig. 3. SHAP identified the most important attributes in the RF model for classifying an instance as 'Dropped Out.' The waterfall plot revealed that attendance percentage and type of residential area (rural) had the greatest impact on the prediction.

impact, reducing the likelihood of dropout. These results highlight the significant role of socioeconomic and academic factors in dropout prediction.

For both classifiers, it was observed that LIME considered the attributes number of failures, gender (male), and attendance percentage to be highly significant in defining a student's situation. The attendance percentage attribute contributed positively by 20% in the RF model and 9% in the DT model to the dropout prediction. Regarding gender, there was a negative contribution of 14% in the RF model and 12% in the DT model.

As mentioned in Section V-A, the best-performing model was generated by the RF. However, through the explainability provided by LIME, it was observed that the model generated by the DT had a slight advantage in the percentage of classifying the sample as 'Dropped Out,' as indicated by the model's prediction probability.

Regarding the impact coefficients of the top 5 attributes on the classification outcome, the sum of the percentages was higher in the RF model. LIME confirmed this trend when explaining other samples, as evidenced in the supplementary material, suggesting that, despite performance differences, the two models share similarities in the probability and impact of key attributes.

LIME identified that attributes such as school attendance, family income, and place of residence had a significant impact on classifying students as dropouts. These findings reinforce the importance of monitoring school attendance and considering socioeconomic factors in intervention planning. Students from rural areas, for example, may face additional challenges, such as lack of transportation.

In Section V-D, we will explore how these factors can be used to develop more effective educational policies.

C. Local and Global Explainability of Models With Shap

SHAP is a technique that provides both local and global explainability. Locally, it explains individual predictions, while globally, it offers insights into the overall behavior of the model across the entire dataset [8], [18].

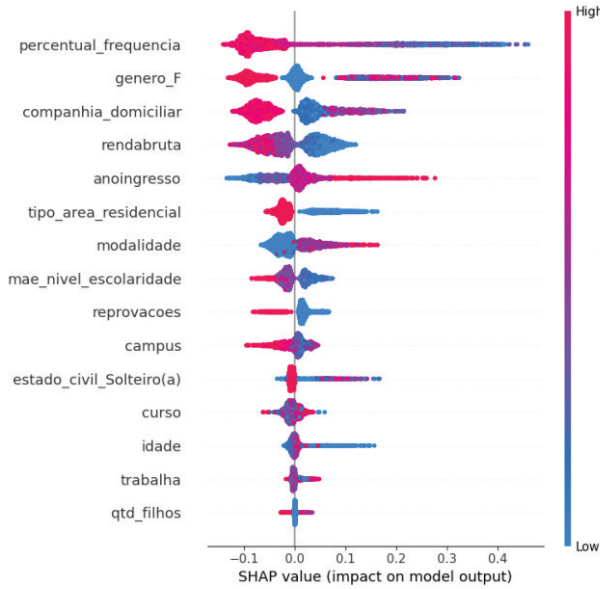


Fig. 4. SHAP (global explanation) in the RF model identified attendance percentage, gender, and gross income as the most relevant attributes for dropout classification, as shown in the summary plot.

Using SHAP's local explanations, the most significant attributes for the classification of an instance were identified. This is evidenced in the waterfall force plot in Fig. 3 for the RF model. This plot allows the visualization of the most influential variables, presented in a cascading manner, along with their percentage contributions that drive the classification towards positive (Dropped Out) or negative (Completed).

The RF model obtained a base value ($E[f(x)] = 0.57$), and from the sum of the positive and negative contributions of the attributes, it reached a prediction probability greater than 82%, as shown in the function $f(x)$ of the plot.

A similar result was obtained when applying SHAP to the DT model to classify an instance as 'Dropped Out,' with a slight increase in prediction probability, which was 83.8%. Again, the attendance percentage had the greatest impact on the prediction (26% positive for dropout), followed by gross income (7% negative), type of residential area (rural) (6% positive), year of enrollment (2% negative), and modality (integrated education) (2% positive).

The model's predictions are positively influenced by points to the right of zero on the horizontal axis, in this case moving closer to the 'Dropped Out' class. Conversely, values to the left of zero on the same axis have a negative impact, tending to drive the prediction toward the 'Completed' class.

By applying SHAP's global explanations to the dataset balanced by the combination of undersampling and oversampling, the most important attributes for classification in the RF and DT models were identified, as shown in Fig. 4 and Fig. 5.

The SHAP summary plots in these figures display the effect of each attribute on the prediction of the evaluated models (the attributes are arranged in descending order, from top to bottom). Each point in the plot represents the `shap_value` calculated for the attribute in a sample.

Thus, the attributes attendance percentage, gender, gross income, and year of enrollment were strongly impactful among

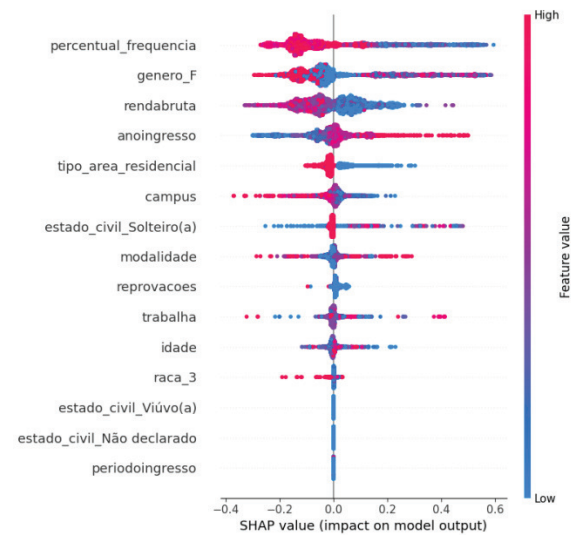


Fig. 5. Main Attributes Identified by SHAP (Global Explanation) in the DT Model. The summary plot highlights relevant factors for dropout classification: attendance percentage, gross income and year of enrollment.

the top 5 attributes in both models. Fig. 4 and Fig. 5 illustrate that the gross income attribute, present in both models, clearly indicates that low values (represented in blue tones) have a positive impact on student dropout. On the other hand, high values (represented in red tones) have a negative impact on dropout, favoring the 'Completed' class.

SHAP confirmed the importance of attributes such as school attendance and family income, corroborating the findings from LIME. This consistency between explainability techniques reinforces the reliability of the model and the relevance of these attributes for dropout prediction.

In Section V-D, we will discuss how this consistency can increase the confidence of educational managers in adopting AI models.

D. Discussion on the Results Obtained for Addressing School Dropout

The XAI methods used in this study were LIME and SHAP, applied to student data to identify the most important attributes for predicting a student's situation. Table VII presents a comparison of the findings of this case study with other studies in literature.

To ensure consistency of the results, cross-validation was applied with LIME and SHAP (graphs in the supplementary material available at <https://bit.ly/case-study-dropout-xai>). The main attributes identified were:

- LIME: Gender, attendance percentage, failures, enrollment period, and gross income;
- SHAP: Attendance percentage, gender, year of enrollment, gross income, and type of residential area.

The recurrence of attributes such as attendance percentage, gross income, and residential areas reinforces their influence on school dropout. The RF model, balanced by SMOTEENN, showed the best performance (AUC and Recall of 0.854), identifying dropout factors like those identified in studies by [6], [8], and [20].

TABLE VII

COMPARISON OF THIS STUDY WITH OTHERS IN THE LITERATURE. LIME AND SHAP WERE APPLIED TO THE RF AND DT MODELS IN THIS STUDY. THIS STUDY REVEALS THAT ATTENDANCE, INCOME, AND RESIDENTIAL AREA SIGNIFICANTLY INFLUENCE DROPOUT. ALTHOUGH LIMITED TO TECHNICAL EDUCATION, THE STUDY INTEGRATES ACADEMIC AND SOCIOECONOMIC FACTORS, WHICH CAN SUPPORT THE DEVELOPMENT OF EDUCATIONAL POLICIES

Aspect	This Study	Study [6]	Study [7]	Study [8]
Objective	Identify dropout indicators in technical education students using XAI	Predict school dropout at IFRN using XAI	Identify at-risk higher education students with a focus on explainability	Identify at-risk students in higher education using data mining
XAI Techniques	LIME and SHAP	LIME and SHAP	LIME, Top-K	LIME, SHAP, and GLOBAL SURROGATE MODEL
ML Models	RF, DT, LR, SVM	Neural Networks	RF, DT, SVM, proposed model	RF
Performance	RF: AUC 0.854, Recall 0.854, DT: AUC 0.818, Recall 0.818	Neural network precision: 99% (training) and 97% (testing)	Weekly evaluation of AUC, precision, recall, and F1-score	Precision of 0.89
Relevant Attributes	Attendance, gross income, modality, residential area	Grades, attendance, family income	Weekly learning activities	GPA, absences, failures
Limitations	Data variability, need for expansion to higher education	Use of a single dataset, lack of evaluation in other contexts	Focus only on weekly activities, without considering external factors	Focuses only on the preparatory year, without considering external factors
Contributions	Identification of socioeconomic and academic factors influencing dropout	Proposal of a new explainability index	Top-K metric to identify at-risk students with high weekly precision	Identification of influential characteristics in academic performance

The prioritization of Recall aims to minimize false negatives, ensuring that most at-risk students are identified, even if false positives increase. The choice between RF and DT depends on the balance between performance and interpretability, with DT being more suitable when transparency for managers is required.

Data balancing was essential to improve the detection of at-risk students, a recurring strategy in the literature [6], [7],

[8], [23], [24], and [34]. However, it may introduce noise, and future research could evaluate alternatives such as ADASYN to assess its impact on model performance.

Table VII shows that LIME and SHAP were used in studies addressing school dropout, such as [6], [7], [8], [19], [20], and [21] indicating that their combination enhances the identification of critical factors, such as school attendance, gross income, gender, and geographic origin. School attendance, identified by LIME as the most impactful factor, reinforces studies like [6], suggesting the need for monitoring and support, such as transportation grants and personalized follow-up.

The consistency between LIME and SHAP validates the robustness of the model, essential for AI applications in education. Compared to [6], it is observed that combining explainability techniques increases the reliability of the analyses. Despite the challenges, the use of SMOTEENN represents an advancement over [6], [7].

In this study, the explainers detected relevant socioeconomic and academic factors, strengthening confidence in the results. Attendance percentage was the main indicator of dropout, while gross income favored course completion, corroborating [6]. Residential area, in line with [20], also proved relevant, especially for students from rural areas, who may face financial and transportation challenges.

Study [35] highlights the impact of income (similarly to [20]) and parental education on dropout. In this study, although parental education is not among the top five attributes, the mother's education level is among the eight most impactful in the RF model (Fig. 4). This difference may be related to factors such as institutional context and methodology adopted.

Thus, preventive actions can be suggested, such as scholarship programs and transportation assistance, already implemented in Brazilian federal institutes. Additionally, in 2024, the government introduced the "Pé-de-Meia" policy, providing financial support to high school students [4].

The findings of this study can be incorporated into institutional policies, enabling early identification of at-risk students and their prioritization in support programs, ensuring greater student retention.

VI. CONCLUSION AND FINAL CONSIDERATIONS

This study applied XAI techniques to identify dropout indicators in a technical education institution, using RF and DT models—a closed-box and a glass-box model, respectively. The models were selected based on predefined criteria, trained on data balanced via undersampling and oversampling techniques, and optimized using the Backward Selection method. The RF model outperformed others, while the DT model ranked fifth.

Both models were analyzed using LIME and SHAP to assess the explainability of key dropout predictors. Despite performance differences, the explanations from both techniques were consistent, identifying school attendance, gross income, and number of failures as the most influential attributes in predicting dropout.

The study recognizes limitations, such as the need for greater data variability, expansion to higher education, broader

comparative analyses, and further contributions to the scientific community. These gaps highlight directions for future research.

In conclusion, this study advances the understanding of AI in education, emphasizing the importance of explainability in machine learning models. The results enhance confidence in decision-making and offer insights to improve student retention and performance.

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